MILITARY SPECIFICATION

EMERGENCY OXYGEN SUPPLY, CHLORATE CANDLE,
AIRCRAFT CRU-74/P

This amendment forms a part of Military Specification
MIL-E-83252 (USAF) dated 20 February 1970.

Page 2, paragraph 2.1, under Military Standards: Delete "MS33660" and the title thereto. Add "MIL-STD-882 System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for".

Page 2: Add:

"2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Society of Automotive Engineers, Inc.

Aerospace Material Specification

AMS 3303 Silicone Rubber, General Purpose, 55 - 65

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)"

Page 3, paragraph 3.4.1.1: Delete and substitute:

"3.4.1.1 Mask connection. The emergency oxygen supply shall incorporate a 0.220 ±.005 inch OD metal tube for connecting the Type MBU-8/P mask hose. The mask connection shall be readily accessible with the housing cap removed."

Page 3, paragraph 3.4.2: Delete and substitute:

"3.4.2 Housing cap. The housing cap shall be designed to retain the Type MBU-8/P oxygen mask by suitable attaching provisions designed to mate with the mask mounting lugs shown in MIL-M-83191. The method of attaching the mask to the housing cap shall permit rapid replacement of the mask and shall preclude separation of the mask from the housing

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cap during the vibration and shock tests of this specification and under a pull of 20 pounds. The housing cap shall provide protection for the mask from moisture, contamination, and damage."

Page 3, paragraph 3.4.2.1, line 1: Delete "roughened" and substitute "designed".

Page 5, paragraph 3.4.3.1: Add: "The configuration of the chlorate candle assembly shall be in accordance with figure 2."

Figure 2: Add the attached figure 2 which forms page 3 of this amendment.

Page 5, paragraph 3.4.4: Delete the first sentence and substitute: "A visual means for determining that the chlorate candle has been ignited, used, or expended shall be incorporated in the base (end opposite the ignition mechanism) of the chlorate candle assembly."

Page 5, paragraph 3.4.5.1: Delete the first sentence and substitute: "The length of the lanyard shall permit removal of the housing cap-mask assembly and displacement of the combination so that between 5 and 7 inches of clear space exists between the housing cap-mask assembly and the housing."

Page 6, paragraph 3.6.1: In lines 2 and 3, delete "4.5 liters" and substitute "4.2 liters". In line 4, delete "2.7 liters" and substitute "2.5 liters".

Page 6, paragraph 3.6.1.1, line 3: Delete "2 seconds" and substitute "2.5 seconds".

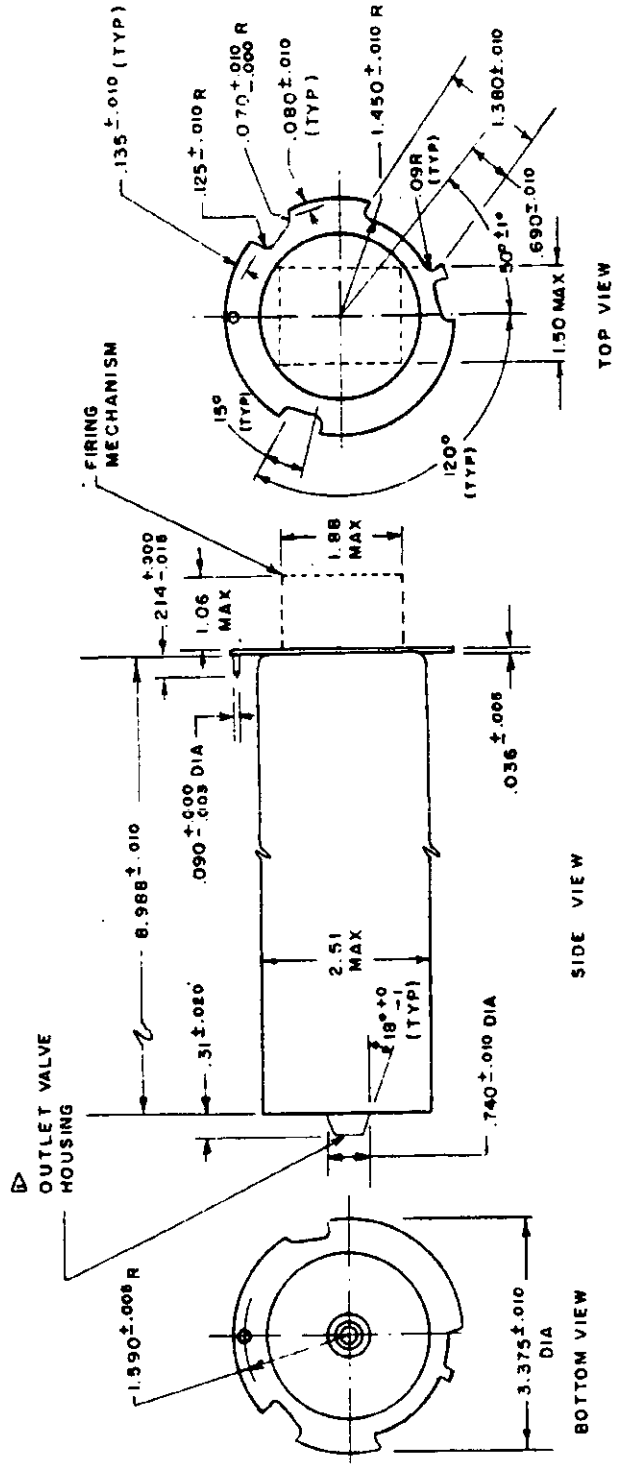
Page 6, paragraph 3.6.5, line 2: Delete "25°F" and substitute "50°F".

Page 7, table I: After the heading of the second column add "3/". At the bottom of the table add the following footnote:

"3/ Unless otherwise specified herein, the limits listed in this column for chlorine, carbon monoxide, and carbon dioxide impurities will be permitted maximum peak tolerances up to 1 part per million by volume, up to 100 parts per million by volume, and up to 20,000 parts per million by volume, respectively. If the limit listed in this column for chlorine, carbon monoxide, or carbon dioxide impurities will be exceeded for a period that will last more than 5 minutes, the maximum peak tolerance for the limit shall not be permitted."

Page 7, paragraph 3.6.7, line 2: Delete "50 psig" and substitute "a value 10 percent below the operating pressure of the relief provision".

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NOTE:
△ MATERIAL : SILICONE 60 SHORE PER AMS 3303
DIMENSIONS IN INCHES

FIGURE 2. Chlorate Candle Assembly Envelope Dimensions

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Page 7, paragraph 3.6.8, line 3: Delete "30 pounds" and substitute "20 pounds".

Page 7, paragraph 3.6.10, lines 2 and 3: Delete "5 pounds" and substitute "8 pounds".

Page 7, paragraph 3.6.11a.: Delete and substitute:

"a. Operating temperatures ranging from +20°F to +120°F and following exposure to storage temperatures ranging from -65°F to +154°F".

Page 8: Add:

"3.13 System safety. A preliminary hazard analysis shall be conducted in accordance with 4.6.16."

Page 10, paragraphs 4.5.2, 4.5.2.1, and 4.5.2.2: Delete and substitute:

"4.5.2 Sampling tests. Two emergency oxygen supplies selected at random from every 100 units, or fraction thereof, shall be subjected to the following tests as described under 4.6:

a. Oxygen flow and flow initiation

b. Actuation device operating pull

c. Oxygen purity

d. Oxygen temperature."

Page 11, paragraph 4.6.2: In line 4, delete "2 seconds" and substitute "2.5 seconds". In line 5, delete "4.5 liters" and substitute "4.2 liters". In line 6, delete "2.7 liters" and substitute "2.5 liters".

Pages 11 and 12, paragraph 4.6.6: In line 5, delete "45°F" and substitute "50°F above ambient temperature". In line 8, delete "120°F" and substitute "145°F".

Page 12, paragraph 4.6.8, lines 1 and 2: Delete "50 psig through the mask attachment fitting" and substitute "a value 10 percent below the operating pressure of the relief provision (in a manner which assures that the test pressure exists throughout the candle assembly and associated connections)".

Page 12, paragraph 4.6.9, line 2: Delete "30 pounds" and substitute "20 pounds".

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Page 12, paragraph 4.6.11, line 5: Delete "5 pounds" (two places) and substitute "8 pounds".

Page 13, paragraphs 4.6.12.7 and 4.6.12.8, line 1: After "oxygen supply" add "with the MBU-8/P mask in place".

Page 14, paragraph 4.6.13: At end of paragraph add: "Acceptance for the reliability test shall be judged on the basis of the accept-reject criteria of test plan I, test level E of MIL-STD-781."

Page 14: Add:

"4.6.15 Mask attachment provision. With the MBU-8/P mask attached to a restrained housing cap, a force of 20 pounds shall be applied to the mask body for a period of 3 seconds in a direction parallel to the plane of the mask mounting lugs and in a manner tending to separate the mask from the housing cap. The mask shall remain attached to the housing cap.

4.6.16 System safety. A preliminary hazard analysis shall be conducted in accordance with MIL-STD-882 to include but not be limited to potential ignition hazards during crash-landing situations with the emergency oxygen supply operating and not operating."

Custodian:
Air Force - 11

Review activities:
Air Force - 71, 84

Preparing activity:
Air Force - 11

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